

09

AS BUILT

N.					
Date					
N.					
Date					
E	N. DUR	HAR	DEG	AS BUILT	ASB
	Date 35/08/09	35/08/09	35/08/09		
D	N. XAL	HAR	DEG	UP DATED TO CEGELEC DESIGN & GOOD FOR EXECUTION	GFE
	Date 94/01/26	94/01/26	94/01/26		
C	N. HFN	HAR	DEG	UP DATED ACCORDING TO EN006 + CEGELEC STUDIES AND MANUFACTURER DATAS	GFE
	Date 93/02/09	93/02/09	93/02/09		
B	N. HUE	HAR	DEG	UP DATED ACCORDING TO CEGELEC STUDIES	APP
	Date 92/12/23	92/12/23	92/12/23		
A	N. HUE	HAR	DEG	ORIGINAL ISSUE	APP
	Date 92/11/13	92/11/13	92/11/13		
W	ETABL	VERIFIE	APPROUVE	MODIFICATIONS	STAT.
	ESTABLISHED	CHECKED	APPROVED		

VINH-SON HYDRO-POWER STATION

2 x 40 MVA

CONTRACT N° 12-05/CEG-PC3-PET/01-91

SOCIALIST REPUBLIC OF VIET-NAM
MINISTRY OF ENERGY
PETECHIM-POWER COMPANY N° 3
ENGINEER EDF



LEADER CEGELEC FRANCE

ALSTHOM - JEUMONT
CEGELECNEYRPC
SDEMECHELLE
SCALE

COMMONS POWER HOUSE
125 VDC SWITCHBOARD
20LAA 001TB
SINGLE LINE DIAGRAM

	DATE	NOM NAME	VISA
ETABL	92/11/13	HUE	
VERIFIE	92/11/13	HAR	
CHECKED	92/11/13	DEG	
APPROUVE	92/11/13	DEG	

CEGELEC

Division ERE P : 35, rue d'Alsace F - 92531 Lavallois Perret Cedex

CGE VIN20 SLAA SU001

P2135A30070-E

FOLIO
SHEET 1
FIN
SHEET 7

AUTC21E1309

1	2	3	4
ANSI	DESIGNATION	CEI	ANSI
12	OVERSPEED DEVICE	n >	50-76
13	SYNCHRONOUS-SPEED DEVICE	n >	50N
14	UNDERSPEED DEVICE	n <	51
15	SPEED MATCHING DEVICE	n <	51
21	DISTANCE OR UNDERIMPEDANCE RELAY	z <	51N
21	DISTANCE OR UNDERREACTANCE RELAY	x <	51V
21	DISTANCE OR OVERADMITTANCE RELAY	y <	
23	TEMPERATURE CONTROL DEVICE	θ	53
25	VOLTAGE COMPARISON DEVICE	u >	55
25	PHASE-ANGLE COMPARISON DEVICE	f >	56
25	FREQUENCY COMPARISON DEVICE	f >	58
25	COUPLING DEVICE	↑	59
25	SYNCHRONIZING OR SYNCHRONISM-CHECK DEVICE	uf ↑	59N
26	THERMOMETER DETECTOR	θ	59-81
27	UNDERVOLTAGE RELAY	u >	60
27	NO-VOLTAGE RELAY	u = 0	60
30	ANNUNCIATOR RELAY	⊙	63
32	DIRECTIONAL ACTIVE POWER RELAY	⊙	64
32	DIRECTIONAL REACTIVE POWER RELAY	⊙	67
32	REVERSE ACTIVE POWER RELAY	⊙	67N
32	REVERSE REACTIVE POWER RELAY	⊙	
37	UNDERCURRENT RELAY	p >	71
37	UNDER POWER RELAY	p >	74
39	MECHANICAL CONDITION MONITOR	p >	78
40	EXCITATION FAULT RELAY	p >	
40	NO-EXCITATION RELAY	p = 0	79
46	PHASE-BALANCE OR REVERSE-PHASE CURRENT RELAY	i >	81
47	VOLTAGE RELAY	u >	87
49	THERMAL RELAY	θ	87N
49	THERMOSTAT	θ	95

5	6	7	8
SWITCH	RESIDUAL CURRENT TRANSFORMER	CAPACITOR	MANUALLY OPERATED
DISCONNECTOR	SHUNT	CAPACITOR VOLTAGE DIVIDER	ELECTROMAGNETIC ACTUATOR
CIRCUIT BREAKER	INDICATING INSTRUMENT	INDUCTOR WINDING	PNEUMATIC OR HYDRAULIC CONTROL
CONTACTOR	RECORDING INSTRUMENT	BATTERY	REVERSER
CONTACTOR POLE (contact closed in the unoperated position)	ENERGY METER	CONVERTER OR MEASURING TRANSDUC.	KEY ALWAYS FREE
SWITCH DISCONNECTOR POLE (on load isolating switch)	IMPULSE COUNTER	RECTIFIER	KEY FREE FOR THE POSITION IN THE DIAGRAM
FUSE	MEASURING RELAY OR RELATED DEVICE	PLUG AND SOCKET	KEY IMPRISONED FOR THE POSITION IN THE DIAGRAM
THERMAL EFFECT MAGNETIC EFFECT	SIGNAL LAMP	ELECTRICAL CONTROL	ISOLATED PHASE BUS DUCT
THREE-WAYS SWITCH	LIMITER	LINK	CABLE
ROTATING MACHINE	DOUBLE DISCONNECTOR	INVERTER	OVERHEAD LINE
TRANSFORMER WITH 2 WINDINGS	SPARKGAP	EARTH MASS	CARRIER LINE TRAP
TRANSFORMER WITH 3 WINDINGS	LIGHTNING ARRESTER	PULSE	PLUG (male)
AUTO-TRANSFORMER	RESISTOR	MECHANICAL INTERLOCK	SOCKET (female)
CURRENT TRANSFORMER	VARIABLE RESISTOR	KEY OPERATED	POWER LINE CARRIER COUPLING DEVICE

20LAA

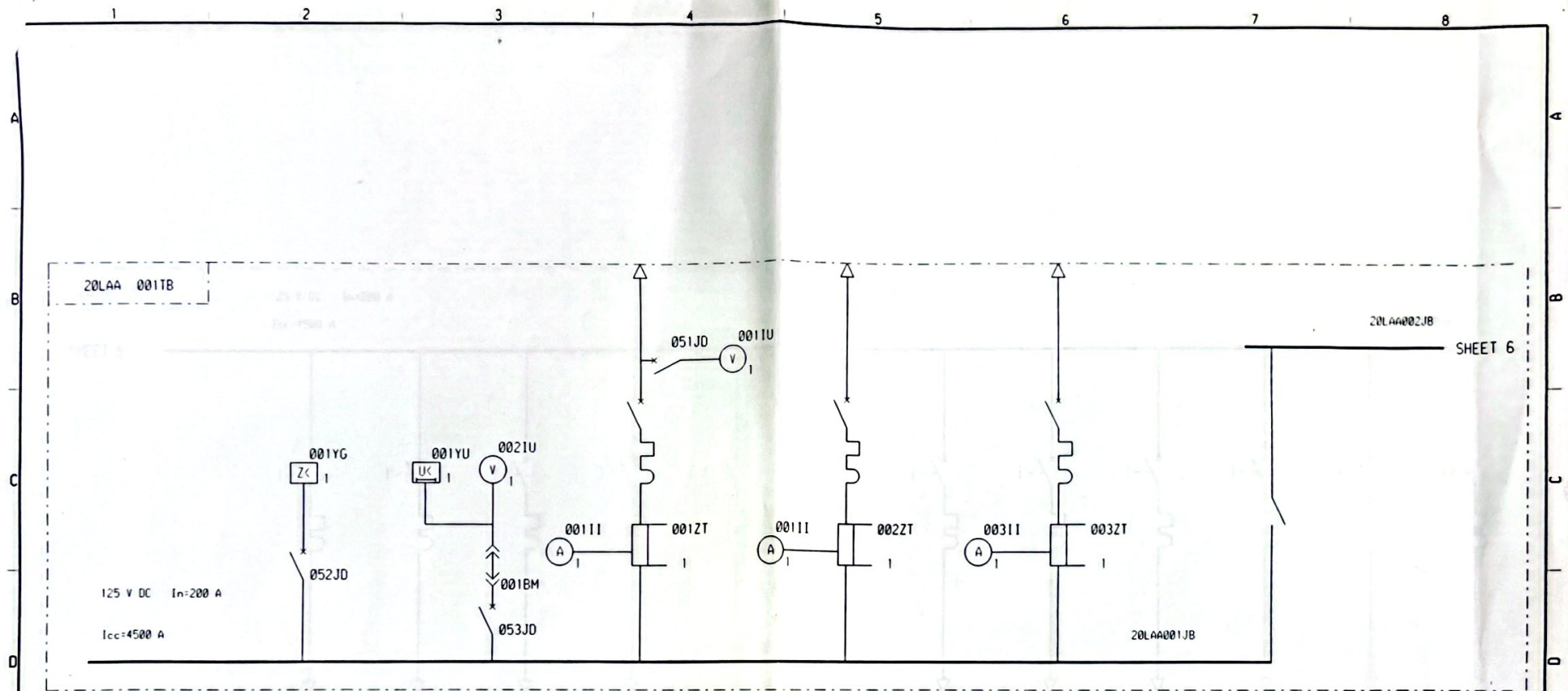
SYMBOLS

P2135A30070 - A

Division
ERE P

3 5

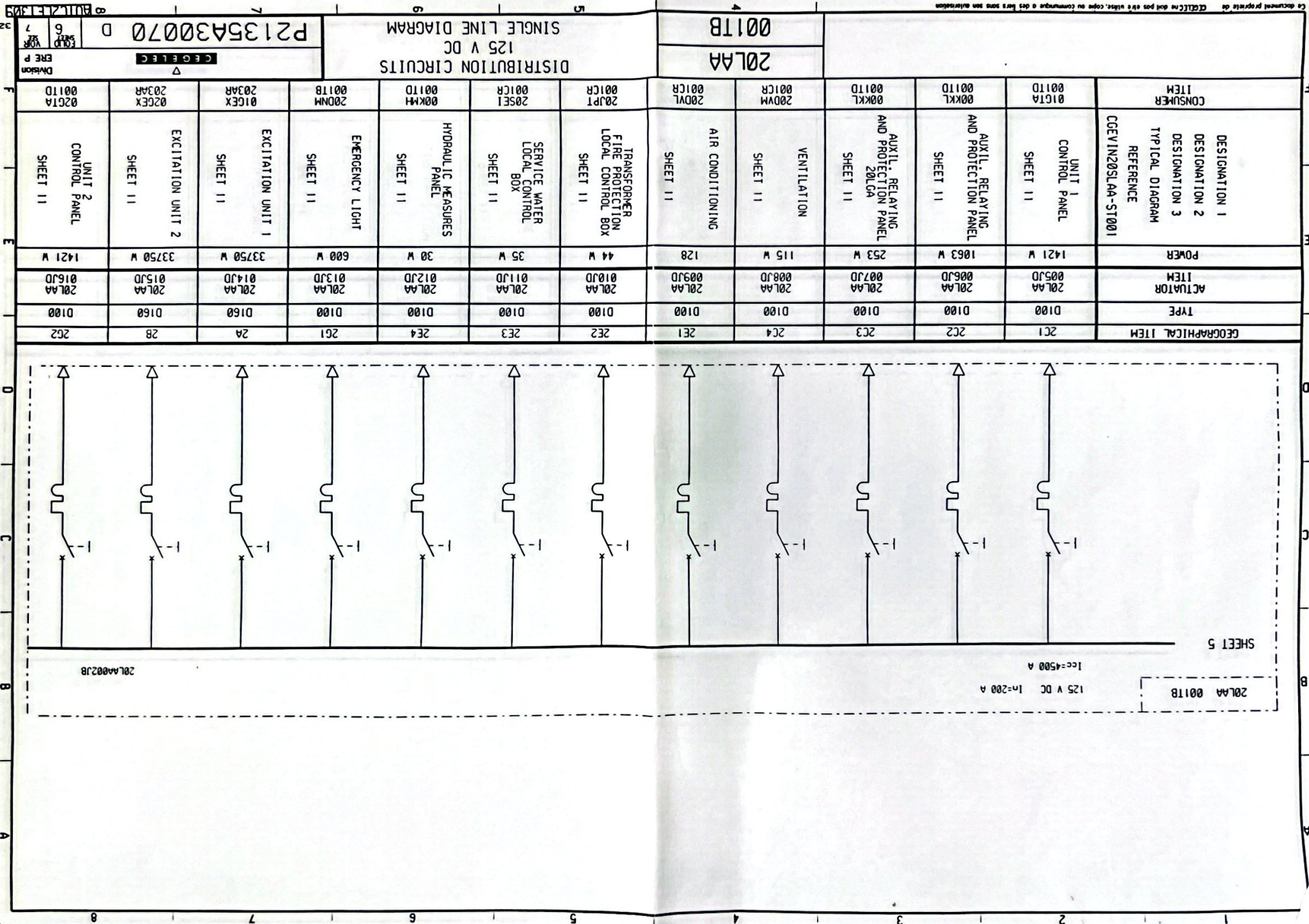
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GEOGRAPHICAL ITEM	01A	01A	01F	01G	01H	01E
TYPE			0250	0250	0250	J250
ACTUATOR ITEM			20LAA 001J0	20LAA 002J0	20LAA 003J0	20LAA 004J1
POWER			400 Ah	18750 W	18750 W	
DESIGNATION 1 DESIGNATION 2 DESIGNATION 3 TYPICAL DIAGRAM REFERENCE CGEV IN 20SLAA-ST001	PERMANENT INSULATED CONTROLLER SHEET 7.14.15.18.19	BUS BARS UNDERVOLTAGE RELAY AND MEASURING SHEET 11	FROM BATTERY 400 Ah 125 V DC SHEET 9	FROM CHARGER RECTIFIER 125 V DC NUMBER 1 SHEET 8	FROM CHARGER RECTIFIER 125 V DC NUMBER 2 SHEET 8	BUS BARS COUPLING SWITCH SHEET 10
CONSUMER ITEM			20LAA 001BT	20LAA 001RD	20LAA 002RD	
			20LAA 001TB	INCOMING FEEDERS 125 V DC SINGLE LINE DIAGRAM		Division ERE P CEGELEC P2135A30070 FOLD SHEET 5 VOR SEE 6

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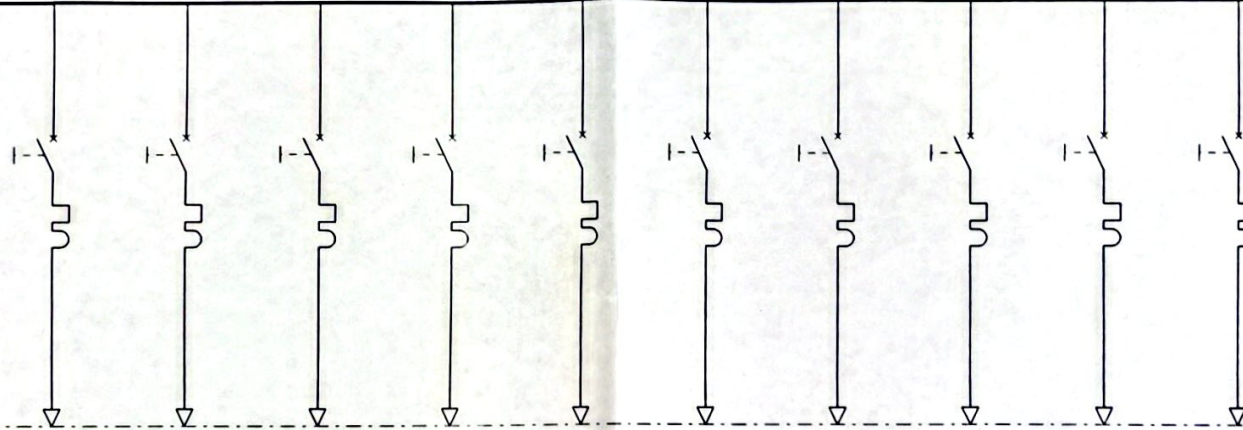
20LAA 001TB

125 V DC In: 200 A

Icc: 4500 A

20LAA00121B

SHEET 6



GEOGRAPHICAL ITEM	2G3	2G4	2I1	2I2	2I3	2I4	2K1	2K2	2K3	2K4		
TYPE	D100	D100	D100	D100	D100	D100	D100	D100	D100	D100		
ACTUATOR ITEM	20LAA 017JD	20LAA 018JD	20LAA 019JD	20LAA 020JD	20LAA 021JD	20LAA 022JD	20LAA 023JD	20LAA 024JD	20LAA 025JD	20LAA 026JD		
POWER	2000 W	2000 W	54 W	136 W	136 W	40 W		44.5 W				
DESIGNATION 1 DESIGNATION 2 DESIGNATION 3 TYPICAL DIAGRAM REFERENCE CCEVIN20SLAA-ST001	INVERTER 1 SHEET 11	INVERTER 2 SHEET 11	SWITCHYARD CONTROL PANEL COUPLING SHEET 11	SWITCHYARD CONTROL PANEL LINE 1 SHEET 11	SWITCHYARD CONTROL PANEL LINE 2 SHEET 11	SWITCHYARD CONTROL PANEL BUS BARS SHEET 11	EQUIPPED SPARE SHEET 11	UNITS FIRE PROTECTION SHEET 11	EQUIPPED SPARE SHEET 11	EQUIPPED SPARE SHEET 11		
CONSUMER ITEM	20LNA 001DL	20LNA 002DL	60LRC 001TD	61LRL 001TD	62LRL 001TD	60LRC 001TD		20JPA 001CR				

20LAA

001TB

DISTRIBUTION CIRCUITS
125 V DC
SINGLE LINE DIAGRAM

P2135A30070

Division
ERE PFOLD
VOR

88 ENCL 21 E 134